Johannes Kepler: Laws of Planetary Motion (1609)

1. The orbit of every planet is an ellipse with the Sun at one of the two foci.

2. A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time.

3. The square of the orbital period of a planet is directly proportional to the cube of the semi-major axis.

Mathematical Formulation: T^2 = k * a^3